

- Description

This module manages a smart greenhouse temperature control system, ensuring that:

1. **The temperature inside the greenhouse is continuously monitored to maintain optimal conditions for plant growth.**
2. **If the temperature exceeds the maximum threshold, the cooling system is activated to lower the temperature.**
3. **If the temperature drops below the minimum threshold, the heating system is activated to raise the temperature.**
4. **A log of all temperature adjustments and system activations is maintained for environmental optimization.**

This system helps improve plant growth efficiency, optimize energy consumption, and automate climate control.

- Specification

module SmartGreenhouseTemperatureControl

var currentTemperature, maxTemperature, minTemperature: real

var coolingActive, heatingActive: boolean

process updateTemperature(changeAmount: real)

functional scenario form:

currentTemperature <= maxTemperature && currentTemperature =
currentTemperature + changeAmount, coolingActive = false, heatingActive = false,
Temperature maintained within optimal range.

||

currentTemperature > maxTemperature && coolingActive = true, heatingActive
= false, Cooling system activated.

||

currentTemperature < minTemperature && coolingActive = false,
heatingActive = true, Heating system activated.

end process

end module

- Java Code

```
package code;
class SmartGreenhouse {
    private double maxTemperature;
    private double minTemperature;
    private double currentTemperature;
    private boolean coolingActive;
    private boolean heatingActive;
    public SmartGreenhouse(double maxTemperature, double minTemperature
, double initialTemperature) {
        this.maxTemperature = maxTemperature;
        this.minTemperature = minTemperature;
    }
}
```

```

        this.currentTemperature = initialTemperature;
        this.coolingActive = false;
        this.heatingActive = false;
    }
    public void updateTemperature(double changeAmount) {
        currentTemperature += changeAmount;
        manageTemperatureControl();
        System.out.printf("Current Temperature: %.2f°C\n", currentTemperature);
        if (coolingActive) {
            System.out.println("Cooling System Activated.");
        }
        if (heatingActive) {
            System.out.println("Heating System Activated.");
        }
    }
    private void manageTemperatureControl() {
        if (currentTemperature > maxTemperature) {
            coolingActive = true;
            heatingActive = false;
        } else if (currentTemperature < minTemperature) {
            coolingActive = false;
            heatingActive = true;
        } else {
            coolingActive = false;
            heatingActive = false;
        }
    }
}

public class c61_SmartGreenhouseSystem {
    public static void main(String[] args) {
        SmartGreenhouse greenhouse = new SmartGreenhouse(30.0, 15.0, 22.0);

        // testcase-VT:
        greenhouse.updateTemperature(10.0);
        greenhouse.updateTemperature(-12.0);
        greenhouse.updateTemperature(-8.0);
        greenhouse.updateTemperature(6.0);

        // testcase-FT:
        // greenhouse.updateTemperature(0.0);
        // greenhouse.updateTemperature(4.9E-324);
        // greenhouse.updateTemperature(-1.7976931348623157E308);
        // greenhouse.updateTemperature(8.389234629357444E307);
    }
}

```

```

        // testcase-Z3:
//      greenhouse.updateTemperature(10.0);
//      greenhouse.updateTemperature(-12.0);
//      greenhouse.updateTemperature(-8.0);
//      greenhouse.updateTemperature(6.0);

        // testcase-UVT:
        greenhouse.updateTemperature(10.0);
        greenhouse.updateTemperature(-12.0);
        greenhouse.updateTemperature(-8.0);
        greenhouse.updateTemperature(6.0);
    }
}

```

- Coverage report

1. Test case number: 4
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c61_SmartGreenhouseSystem.java	100.0 %	8	0	8
> c61_SmartGreenhouseSystem		0	0	0
> SmartGreenhouse	100.0 %	8	0	8

3. Vibration testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	104	20,604	20,708
▼ src	0.5 %	104	20,604	20,708
▼ code	0.5 %	104	20,604	20,708
▼ c61_SmartGreenhouseSystem.java	97.2 %	104	3	107
> c61_SmartGreenhouseSystem	87.0 %	20	3	23
> SmartGreenhouse	100.0 %	84	0	84

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c61_SmartGreenhouseSystem.java	75.0 %	6	2	8
> c61_SmartGreenhouseSystem		0	0	0
> SmartGreenhouse	75.0 %	6	2	8

5. Fuzz testing instruction coverage: 87.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	94	20,614	20,708
▼ src	0.5 %	94	20,614	20,708
▼ code	0.5 %	94	20,614	20,708
▼ c61_SmartGreenhouseSystem.java	87.9 %	94	13	107
> c61_SmartGreenhouseSystem	87.0 %	20	3	23
> SmartGreenhouse	88.1 %	74	10	84

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c61_SmartGreenhouseSystem.java	100.0 %	8	0	8
> c61_SmartGreenhouseSystem		0	0	0
> SmartGreenhouse	100.0 %	8	0	8

7. Z3 instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	104	20,604	20,708
▼ src	0.5 %	104	20,604	20,708
▼ code	0.5 %	104	20,604	20,708
▼ c61_SmartGreenhouseSystem.java	97.2 %	104	3	107
> c61_SmartGreenhouseSystem	87.0 %	20	3	23
> SmartGreenhouse	100.0 %	84	0	84

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c61_SmartGreenhouseSystem.java	100.0 %	8	0	8
> c61_SmartGreenhouseSystem		0	0	0
> SmartGreenhouse	100.0 %	8	0	8

9. Unimproved V-method instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	104	20,604	20,708
▼ src	0.5 %	104	20,604	20,708
▼ code	0.5 %	104	20,604	20,708
▼ c61_SmartGreenhouseSystem.java	97.2 %	104	3	107
> c61_SmartGreenhouseSystem	87.0 %	20	3	23
> SmartGreenhouse	100.0 %	84	0	84